POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6022 | Semester VI | Theory | 4 Credits

Mechatronics and Automation

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6022 as Mechatronics and Automation in Semester VI for Mechanical Engineering. The official SITTTR detailed course page was unavailable. With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Guwahati’s Automation in Manufacturing outline and NPTEL IIT Roorkee’s Mechatronics course description. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Wiring, PLC programming, hydraulic/pneumatic work, machine motion, CNC operation and troubleshooting must use authorised equipment, documented risk controls, isolation/lockout and competent supervision.

Verified source note: Verified sources support mechatronic-system concepts, automated-system design, sensors/transducers, signal conditioning, microprocessors, electric drives, mechanisms, hydraulic/pneumatic systems and CNC technology. The four-module study sequence is not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Mechatronics and Automation studies the purposeful integration of mechanical systems, electronics, sensing, actuation, computation and control. It develops the systems view needed to understand industrial automation, from a physical process and its sensors through logic, drives, mechanisms, fluid power, handling and CNC.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□□ □□□□□□□□□ □□□□□□□□□□.

Prerequisite foundation

Basic electrical/electronic principles, mechanics, manufacturing processes, engineering drawing, computer awareness and strict laboratory/workshop safety.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain mechatronics, automation levels and the building blocks of an automated system.
2. Explain sensors, transducers, signal conditioning and data/interface concepts.
3. Explain actuators, electric drives, mechanisms, hydraulic and pneumatic systems.
4. Explain programmable control, industrial automation, CNC and safe system-integration principles.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO ▯▯▯▯▯▯ exam-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯ ▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯. Define, explain, apply ▯▯▯▯▯ action words ▯▯▯▯▯▯▯▯▯▯▯▯.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the interdisciplinary structure of a mechatronic system and the purpose of industrial automation.	Understand
CO2	Explain sensing, transduction, signal conditioning and selection of measurement elements.	Understand
CO3	Explain electrical, mechanical, hydraulic and pneumatic actuation elements used in automation.	Understand
CO4	Explain programmable control, CNC and integrated automation with safety, quality and maintenance awareness.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Mechatronic Systems and Industrial Automation	Definition and scope of mechatronics; open/closed-loop systems; automation types; system blocks; design requirements, interfaces and life-cycle thinking.	Approx. 14 hours	CO1	Module 1
2	Sensors, Transducers and Signal Interfaces	Measurement-system concepts; displacement, proximity, temperature, pressure and flow sensing; static performance; signal conditioning, conversion, protection and selection.	Approx. 15 hours	CO2	Module 2
3	Actuation, Drives, Mechanisms and Fluid Power	DC/AC, stepper and servo drives; motion transmission and mechanisms; hydraulic pumps/valves; pneumatic supply, treatment, regulation and circuits.	Approx. 16 hours	CO3	Module 3
4	Programmable Control, CNC and Integrated Practice	Microprocessor/microcontroller and PLC concepts; input–logic–output flow; sequencing and diagnostics; CNC/CAM awareness; integrated automation, safety, quality and maintenance.	Approx. 15 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Mechatronic Systems and Industrial Automation

Definition and scope of mechatronics; open/closed-loop systems; automation types; system blocks; design requirements, interfaces and life-cycle thinking.

CO1 · Approx. 14 hours

Sensors, Transducers and Signal Interfaces

Measurement-system concepts; displacement, proximity, temperature, pressure and flow sensing; static performance; signal conditioning, conversion, protection and selection.

CO2 · Approx. 15 hours

Actuation, Drives, Mechanisms and Fluid Power

DC/AC, stepper and servo drives; motion transmission and mechanisms; hydraulic pumps/valves; pneumatic supply, treatment, regulation and circuits.

CO3 · Approx. 16 hours

Programmable Control, CNC and Integrated Practice

Microprocessor/microcontroller and PLC concepts; input–logic–output flow; sequencing and diagnostics; CNC/CAM awareness; integrated automation, safety, quality and maintenance.

CO4 · Approx. 15 hours

MODULE 1

Mechatronic Systems and Industrial Automation

Definition and scope of mechatronics; open/closed-loop systems; automation types; system blocks; design requirements, interfaces and life-cycle thinking.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

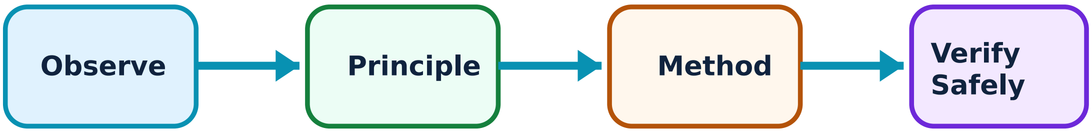
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Mechatronic Systems and Industrial Automation: identify the system, state its principle, follow the correct method and verify the result safely.

1. The mechatronic systems view−

Mechatronics integrates mechanics with electronics and intelligent computer control. A typical system contains a physical process, sensors, signal/interface elements, controller, actuators/drives, feedback and a human/machine interface.

Study and application method

Draw a block diagram for an automated mechanism and label the energy path, information path, feedback path, disturbance and safety function.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a block diagram for an automated mechanism and label the energy path, information path, feedback path, disturbance and safety function.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A block diagram is an analytical model, not a safe wiring or control design. Safety functions must be engineered independently and validated for the application.

2. Automation purpose and levels−

Automation can improve repeatability, productivity, quality, ergonomics and traceability, but its suitability depends on product variation, volume, investment, skills, reliability and maintainability. Fixed, programmable and flexible automation have different uses.

Study and application method

Compare alternative automation levels for a stated production case, including expected benefits, constraints, human role, quality checks and fallback/maintenance needs.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Do not frame automation as replacing all human judgement; safe operation, inspection, escalation and accountable oversight remain essential.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Sensors, Transducers and Signal Interfaces

Measurement-system concepts; displacement, proximity, temperature, pressure and flow sensing; static performance; signal conditioning, conversion, protection and selection.

Approx. 15 hours | **CO2** | **Understand · Apply**

Learning goals

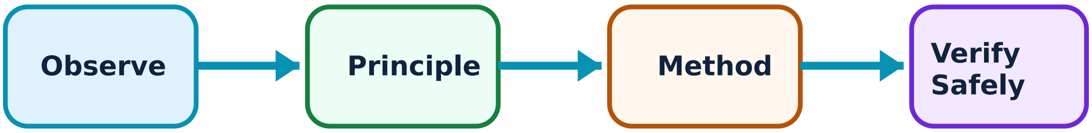
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Sensors, Transducers and Signal Interfaces: identify the system, state its principle, follow the correct method and verify the result safely.

1. Sensing and measurement characteristics−

Sensors/transducers convert a physical variable into a usable signal. Selection considers range, accuracy, resolution, sensitivity, repeatability, response time, environmental resistance, mounting, calibration and signal compatibility.

Study and application method

Prepare a sensor-selection table for one variable, stating measurand, range, required accuracy, environment, output type, installation constraint and the evidence needed to confirm suitability.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a sensor-selection table for one variable, stating measurand, range, required accuracy, environment, output type, installation constraint and the evidence needed to confirm suitability.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Sensor outputs can be misleading when wiring, mounting, calibration, EMI, contamination or reference conditions are wrong; verify using an approved procedure.

2. Signal conditioning and interfaces−

Amplification, filtering, bridge circuits, protection, analogue-to-digital conversion and isolation can make sensor signals reliable and compatible with control hardware. Data quality affects every downstream control decision.

Study and application method

Trace a signal from sensor to controller input, identifying conditioning, conversion, common noise/error sources and one diagnostic check for each stage.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Trace a signal from sensor to controller input, identifying conditioning, conversion, common noise/error sources and one diagnostic check for each stage.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not connect unknown signals directly to a controller or measurement device. Confirm voltage/current ranges, grounding, isolation and manufacturer requirements first.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Actuation, Drives, Mechanisms and Fluid Power

DC/AC, stepper and servo drives; motion transmission and mechanisms; hydraulic pumps/valves; pneumatic supply, treatment, regulation and circuits.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

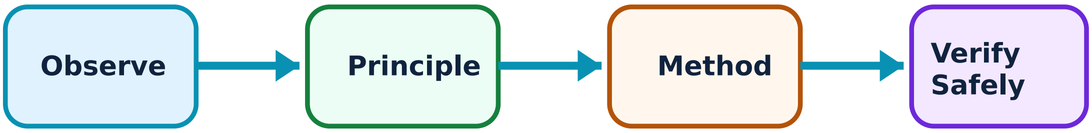
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Actuation, Drives, Mechanisms and Fluid Power: identify the system, state its principle, follow the correct method and verify the result safely.

1. Electrical drives and mechanisms−

Motors and drives turn control commands into motion. DC/AC motors, steppers and servos differ in control, torque-speed behaviour, positioning and feedback. Ball screws, cams, indexing and handling mechanisms transform rotary or linear motion for automated tasks.

Study and application method

For an axis or handling task, identify required motion, load, duty cycle, accuracy, feedback and mechanism; then justify a provisional drive/mechanism choice.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For an axis or handling task, identify required motion, load, duty cycle, accuracy, feedback and mechanism; then justify a provisional drive/mechanism choice.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Unexpected motion can cause serious injury. Design and test only with guards, emergency-stop strategy, safe speed/torque limits and authorised supervision.

2. Hydraulic and pneumatic actuation−

Hydraulics use pressurised fluid for high-force compact actuation; pneumatics use compressed air for clean, rapid and often simpler motion. Pumps/compressors, valves, regulators, cylinders and graphical circuit symbols support system description.

Study and application method

Read a simple circuit diagram to identify supply, directional control, flow/pressure control, actuator and likely sequence. Explain the function without changing any physical connection.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Read a simple circuit diagram to identify supply, directional control, flow/pressure control, actuator and likely sequence. Explain the function without changing any physical connection.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Stored pressure, hose failure, fluid injection, flying parts and noise are hazards. Depressurise/isolate only through the approved procedure and never locate leaks by hand.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Programmable Control, CNC and Integrated Practice

Microprocessor/microcontroller and PLC concepts; input–logic–output flow; sequencing and diagnostics; CNC/CAM awareness; integrated automation, safety, quality and maintenance.

Approx. 15 hours | CO4 | Understand · Apply

Learning goals

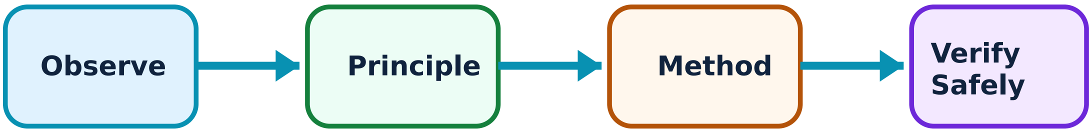
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Programmable Control, CNC and Integrated Practice: identify the system, state its principle, follow the correct method and verify the result safely.

1. Programmable control and sequencing−

Controllers receive input states, execute programmed logic and command outputs. PLCs are widely used in industrial automation because they support robust I/O, diagnostics and maintainable logic for sequences, interlocks and timing.

Study and application method

Express a simple safe sequence as a state table: preconditions, inputs, outputs, transition condition, fault response and reset condition. Use simulation or instructor-approved hardware only.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Express a simple safe sequence as a state table: preconditions, inputs, outputs, transition condition, fault response and reset condition. Use simulation or instructor-approved hardware only.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: A functioning sequence is not automatically safe. Interlocks, safety circuits, mode control, restart prevention and validation need application-specific engineering.

2. CNC and integrated automation−

CNC combines programmed motion with machine control; automated manufacturing may link material handling, sensing, tooling, process planning, quality data and operator interfaces. Integration requires compatibility, commissioning discipline and maintainability.

Study and application method

Map an integrated cell from order/part data to setup, machining/handling, inspection, traceability and maintenance feedback. Identify at least one interface risk and its verification method.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map an integrated cell from order/part data to setup, machining/handling, inspection, traceability and maintenance feedback. Identify at least one interface risk and its verification method.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: CNC and automated cells require access control, guarding, verified programs, tool/fixture checks and documented change management; never run unverified code on a live machine.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Mechatronic Systems and Industrial Automation

Use the concept flow to revise observation, principle, method and verification.

Module 2: Sensors, Transducers and Signal Interfaces

Use the concept flow to revise observation, principle, method and verification.

Module 3: Actuation, Drives, Mechanisms and Fluid Power

Use the concept flow to revise observation, principle, method and verification.

Module 4: Programmable Control, CNC and Integrated Practice

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Create a block diagram of a small automated mechanism, separating energy, information, control and safety paths.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Complete a sensor-selection and signal-path worksheet for an industrial measurement task.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Interpret supplied hydraulic/pneumatic circuit symbols and explain the intended motion sequence without operating equipment.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Develop a PLC-style state table for an instructor-approved simulated sequence, including faults and reset conditions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare an integration checklist for a CNC/handling cell covering interfaces, inspection, maintenance and safety validation.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Mechatronic Systems and Industrial Automation

Explain The mechatronic systems view and state one correct method, application or precaution.−

Mechatronics integrates mechanics with electronics and intelligent computer control. A typical system contains a physical process, sensors, signal/interface elements, controller, actuators/drives, feedback and a human/machine interface.

Answer framework: Draw a block diagram for an automated mechanism and label the energy path, information path, feedback path, disturbance and safety function.

Important point: A block diagram is an analytical model, not a safe wiring or control design. Safety functions must be engineered independently and validated for the application.

Explain Automation purpose and levels and state one correct method, application or precaution.−

Automation can improve repeatability, productivity, quality, ergonomics and traceability, but its suitability depends on product variation, volume, investment, skills, reliability and maintainability. Fixed, programmable and flexible automation have different uses.

Answer framework: Compare alternative automation levels for a stated production case, including expected benefits, constraints, human role, quality checks and fallback/maintenance needs.

Important point: Do not frame automation as replacing all human judgement; safe operation, inspection, escalation and accountable oversight remain essential.

Module 2 — Sensors, Transducers and Signal Interfaces

Explain Sensing and measurement characteristics and state one correct

method, application or precaution.−

Sensors/transducers convert a physical variable into a usable signal. Selection considers range, accuracy, resolution, sensitivity, repeatability, response time, environmental resistance, mounting, calibration and signal compatibility.

Answer framework: Prepare a sensor-selection table for one variable, stating measurand, range, required accuracy, environment, output type, installation constraint and the evidence needed to confirm suitability.

Important point: Sensor outputs can be misleading when wiring, mounting, calibration, EMI, contamination or reference conditions are wrong; verify using an approved procedure.

Explain Signal conditioning and interfaces and state one correct method, application or precaution.−

Amplification, filtering, bridge circuits, protection, analogue-to-digital conversion and isolation can make sensor signals reliable and compatible with control hardware. Data quality affects every downstream control decision.

Answer framework: Trace a signal from sensor to controller input, identifying conditioning, conversion, common noise/error sources and one diagnostic check for each stage.

Important point: Do not connect unknown signals directly to a controller or measurement device. Confirm voltage/current ranges, grounding, isolation and manufacturer requirements first.

Module 3 — Actuation, Drives, Mechanisms and Fluid Power

Explain Electrical drives and mechanisms and state one correct method, application or precaution.−

Motors and drives turn control commands into motion. DC/AC motors, steppers and servos differ in control, torque-speed behaviour, positioning and feedback. Ball screws, cams, indexing and handling mechanisms transform rotary or linear motion for automated tasks.

Answer framework: For an axis or handling task, identify required motion, load, duty cycle, accuracy, feedback and mechanism; then justify a provisional drive/mechanism choice.

Important point: Unexpected motion can cause serious injury. Design and test only with guards, emergency-stop strategy, safe speed/torque limits and authorised supervision.

Explain Hydraulic and pneumatic actuation and state one correct method, application or precaution.−

Hydraulics use pressurised fluid for high-force compact actuation; pneumatics use compressed air for clean, rapid and often simpler motion. Pumps/compressors, valves, regulators, cylinders and graphical circuit symbols support system description.

Answer framework: Read a simple circuit diagram to identify supply, directional control, flow/pressure control, actuator and likely sequence. Explain the function without changing any physical connection.

Important point: Stored pressure, hose failure, fluid injection, flying parts and noise are hazards. Depressurise/isolate only through the approved procedure and never locate leaks by hand.

Module 4 — Programmable Control, CNC and Integrated Practice

Explain Programmable control and sequencing and state one correct method, application or precaution.−

Controllers receive input states, execute programmed logic and command outputs. PLCs are widely used in industrial automation because they support robust I/O, diagnostics and maintainable logic for sequences, interlocks and timing.

Answer framework: Express a simple safe sequence as a state table: preconditions, inputs, outputs, transition condition, fault response and reset condition. Use simulation or instructor-approved hardware only.

Important point: A functioning sequence is not automatically safe. Interlocks, safety circuits, mode control, restart prevention and validation need application-specific engineering.

Explain CNC and integrated automation and state one correct method, application or precaution.−

CNC combines programmed motion with machine control; automated manufacturing may link material handling, sensing, tooling, process planning, quality data and operator interfaces. Integration requires compatibility, commissioning discipline and maintainability.

Answer framework: Map an integrated cell from order/part data to setup, machining/handling, inspection, traceability and maintenance feedback. Identify at least one interface risk and its verification method.

Important point: CNC and automated cells require access control, guarding, verified programs, tool/fixture checks and documented change management; never run unverified code on a live machine.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Mechatronic Systems and Industrial Automation.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Sensors, Transducers and Signal Interfaces.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Actuation, Drives, Mechanisms and Fluid Power.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Programmable Control, CNC and Integrated Practice.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Definition and scope of mechatronics; open/closed-loop systems; automation types; system blocks; design requirements, interfaces and life-cycle thinking..
2. Develop a complete answer or practical plan for: Measurement-system concepts; displacement, proximity, temperature, pressure and flow sensing; static performance; signal conditioning, conversion, protection and selection..
3. Develop a complete answer or practical plan for: DC/AC, stepper and servo drives; motion transmission and mechanisms; hydraulic pumps/valves; pneumatic supply, treatment, regulation and circuits..
4. Develop a complete answer or practical plan for: Microprocessor/microcontroller and PLC concepts; input–logic–output flow; sequencing and diagnostics; CNC/CAM awareness; integrated automation, safety, quality and maintenance..

LAST-MILE REVISION

Quick Revision

Module 1: Mechatronic Systems and Industrial Automation

Definition and scope of mechatronics; open/closed-loop systems; automation types; system blocks; design requirements, interfaces and life-cycle thinking.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Sensors, Transducers and Signal Interfaces

Measurement-system concepts; displacement, proximity, temperature, pressure and flow sensing; static performance; signal conditioning, conversion, protection and selection.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Actuation, Drives, Mechanisms and Fluid Power

DC/AC, stepper and servo drives; motion transmission and mechanisms; hydraulic pumps/valves; pneumatic supply, treatment, regulation and circuits.

- Match each topic to CO3.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 4: Programmable Control, CNC and Integrated Practice

Microprocessor/microcontroller and PLC concepts; input–logic–output flow; sequencing and diagnostics; CNC/CAM awareness; integrated automation, safety, quality and maintenance.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
The mechatronic systems view	Mechatronics integrates mechanics with electronics and intelligent computer control.
Automation purpose and levels	Automation can improve repeatability, productivity, quality, ergonomics and traceability, but its suitability depends on product variation, volume, investment, skills, reliability and maintainability.
Sensing and measurement characteristics	Sensors/transducers convert a physical variable into a usable signal.
Signal conditioning and interfaces	Amplification, filtering, bridge circuits, protection, analogue-to-digital conversion and isolation can make sensor signals reliable and compatible with control hardware.
Electrical drives and mechanisms	Motors and drives turn control commands into motion.
Hydraulic and pneumatic actuation	Hydraulics use pressurised fluid for high-force compact actuation; pneumatics use compressed air for clean, rapid and often simpler motion.
Programmable control and sequencing	Controllers receive input states, execute programmed logic and command outputs.
CNC and integrated automation	CNC combines programmed motion with machine control; automated manufacturing may link material handling, sensing, tooling, process planning, quality data and operator interfaces.

LEARNING RESOURCES

References

Recommended mechatronics and automation references

1. W. Bolton, Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering.
2. D. Alciatore and M. Histand, Introduction to Mechatronics and Measurement Systems.
3. M. P. Groover, Automation, Production Systems, and Computer-Integrated Manufacturing.
4. S. R. Majumdar, Oil Hydraulic Systems: Principles and Maintenance.

Approved public mechatronics and automation sources

- https://nptel.ac.in/courses/112103293
- https://onlinecourses.nptel.ac.in/noc21_me27/preview

These sources provide the verified study basis while official Course 6022 detail is unavailable; they do not replace machine manuals, electrical standards, safety procedures or authorised control-system documentation.